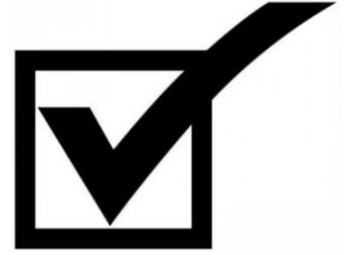




**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential



ADVANCED SPREADSHEET APPLICATIONS

(235)

NATIONAL 2022

Production:

Job 1: Worksheets with answers and formulas	_____ (295 points)
Job 2: Formatted worksheet	_____ (115 points)
Job 3: Clustered column chart	_____ (35 points)
TOTAL POINTS	_____ (445 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. Put your Member ID and printout number in the right section of the footer on each printout unless instructed otherwise. Your name or initials should *not* appear on any work you submit.
5. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
6. When turning in your contest, the jobs should be arranged in printout order.

SCENARIO:

You work for Handy Handyman construction company. Your company has done a complete bathroom remodel and switching of a tub/shower combo to just a shower for Brandy McClain. Today you have to complete the bathroom remodel invoice summary for Brandy McClain with the total costs, which includes how much she paid with her down payment. You need to create and input the information into the invoice spreadsheet, compute the costs, and send to her by email. Then you need to create another worksheet that shows the bathroom remodel comparisons from actual cost to invoice costs, it must show where it cost you more money than you invoiced and if you lost any money on the job. Based on the comparison create a chart.

Job 1 (Worksheets with answers and formulas)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Calibri Size 11 font used for data	10	
Landscape orientation with narrow margins	10	
Worksheet printed landscape on one page centered vertically and horizontally	10	
Type in A1 Handy Handyman	5	
Type in A2 Remodeling Invoice	5	
Type in A3 Items	5	
In B43:C43 a function that calculates the total material costs	10	
Copy A1:B3 to the “Bathroom Remodel Summary” worksheet	5	
Enter a function in A5 that references cell A5 from the original worksheet	5	
Enter a function in A6 that references cell A15 from the original worksheet	5	
Enter a function in A7 that references cell A18 from the original worksheet	5	
Enter a function in A8 that references cell A21 from the original worksheet	5	
Enter a function in A9 that references cell A25 from the original worksheet	5	
Enter a function in A10 that references cell A29 from the original worksheet	5	
Enter a function in A11 that references cell A33 from the original worksheet	5	
Enter a function in A12 that references cell A36 from the original worksheet	5	
Enter a function in A13 that references cell A39 from the original worksheet	5	
Enter a function in B5 that calculates the sum of the Bath_Shower range	5	
Enter a function in B6 that calculates the sum of the Cabinets range	5	
Enter a function in B7 that calculates the sum of the Countertops range	5	
Enter a function in B8 that calculates the sum of the Faucets range	5	
Enter a function in B9 that calculates the sum of the Flooring range	5	
Enter a function in B10 that calculates the sum of the Lighting range	5	
Enter a function in B11 that calculates the sum of the Sinks range	5	
Enter a function in B12 that calculates the sum of the Toilets range	5	
Enter a function in B13 that calculates the sum of the Walls range	5	
Add title and enter a formula in C15 that that totals the labor and materials costs	10	
Add title and enter an absolute formula in C16 that calculates the incidental costs	10	
Add title and enter a formula in C17 for the Job Total	10	
Add title and enter a formula in C19 that calculates the 20% down payment	10	
Add title and enter a formula in C20 that calculates the total due	10	
Copy the worksheet	5	

Delete rows 15-20	5	
Update worksheet with titles and Actual Cost	10	
Enter in formula in D5:D13 that calculates the difference between the total and actual costs	10	
Enter function to calculate the totals in B15:D15	5	
In B15 place a function that totals the total cost column and fill this function over to C15 and D15	10	
In D16 use a reference function to pull the total incidentals information	5	
In D17, insert a formula that calculates the Job Loss/Job Covered by the amount budgeted for incidentals	5	
In cell D18 place a function that finds the highest difference of loss	5	
In cell E3, add the title, "Over/Under". Add a nested If statement in E5 that calculates if the items in over or under budget. Copy the If statement from E5 to E13	20	
Print a copy of the worksheet with gridlines and adjust column widths to make all data visible and print with the formulas showing	5	
Subtotal	295	
Job 2 (Formatted worksheet)	Points Possible	Points Earned
Member ID and Job # in right section of footer	10	
Original Sheet: Merge and center the title A1 through C1.	5	
Merge and center the subtitle A2 through C2.	5	
Format the company name to Title cell style.	5	
Format the subtitle to Heading 4 cell style.	5	
Center and Bold A3:C3	5	
Format columns B and C to comma with 2 decimal places.	5	
Increase column width of all columns using auto fit.	5	
Bathroom Remodel: Merge and center the title A1 through C1.	5	
Merge and center the subtitle A2 through C2.	5	
Format the company name to Title cell style.	5	
Format the subtitle to Heading 4 cell style.	5	
Center and Bold A3:B3	5	
Format B5:C20 to comma with 2 decimal places.	5	
Comparison: Merge and center the title A1 through E1	5	
Merge and center the title A1 through E1	5	
Merge and center the subtitle A2 through E2	5	
Format the company name to Title cell style	5	
Format the subtitle to Heading 4 cell style	5	
Center and Bold A3:E3	5	

Format B5:D18 to comma with 2 decimal places	5	
Print each spreadsheet with all data visible, centered, fit to one page	5	
Subtotal	115	
Job 3 (Clustered Column Chart)	Points Possible	Points Earned
Create a clustered column chart	10	
Select the chart style 2	5	
Title of the chart is "Cost Comparison"	5	
Title is Impact font size 28.	5	
Make sure the legend is showing at the top with data labels at the outside end	5	
Print the chart on a separate sheet	5	
Subtotal	35	
TOTAL POINTS	445	

JOB 1

Handy Handyman		
Remodeling Invoice		
Items	Material Cost	Labor Cost
Bath/Shower		
Tile	400	700
Grouting	50	100
Greenboard	100	150
Shower Pan	500	750
Floor Leveling	80	120
Bench Materials	350	500
Plumbing	200	600
Tub Removal	0	70
Cabinets		
Refinish and Paint	75	250
Countertops		
Granite	350	100
Faucets		
Showers	280	30
Sinks	1200	320
Flooring		
Tile	380	425
Grout	114	220
Lighting		
Above Sinks	160	100
Ceiling Lights	80	180
Sinks		
Undermount	280	600
Toilets		
Installation	375	600
Walls		
Paint	175	760
Subtotal	=SUM(B6:B42)	=SUM(C6:C42)

Handy Handyman		
Remodeling Invoice		
Items	Total Cost	
=Original Spreadsheet!A5	=SUM(Bath_Shower)	
=Original Spreadsheet!A15	=SUM(Cabinets)	
=Original Spreadsheet!A18	=SUM(Countertops)	
=Original Spreadsheet!A21	=SUM(Faucets)	
=Original Spreadsheet!A25	=SUM(Flooring)	
=Original Spreadsheet!A29	=SUM(Lighting)	
=Original Spreadsheet!A33	=SUM(Sinks)	
=Original Spreadsheet!A36	=SUM(Toilets)	
=Original Spreadsheet!A39	=SUM(Walls)	
Labor and Materials		=SUM('Original Spreads
Incidentals	0.1	=C15*\$B\$16
Job Total		=SUM(C15:C16)
Down Payment	0.2	=C17*\$B\$19
Total Due		=C17-C19

Handy Handyman				
Remodeling Invoice				
Items	Total Cost	Actual Cost	Difference	Over/Under
=Original Spreadsheet!A5	=SUM(Bath_Shower)	4585	=B5-C5	=IF(D5>0,"Under",IF(D5=0,"Equal",IF(D5<0,"Over")))
=Original Spreadsheet!A15	=SUM(Cabinets)	630	=B6-C6	=IF(D6>0,"Under",IF(D6=0,"Equal",IF(D6<0,"Over")))
=Original Spreadsheet!A18	=SUM(Countertops)	450	=B7-C7	=IF(D7>0,"Under",IF(D7=0,"Equal",IF(D7<0,"Over")))
=Original Spreadsheet!A21	=SUM(Faucets)	1500	=B8-C8	=IF(D8>0,"Under",IF(D8=0,"Equal",IF(D8<0,"Over")))
=Original Spreadsheet!A25	=SUM(Flooring)	1380	=B9-C9	=IF(D9>0,"Under",IF(D9=0,"Equal",IF(D9<0,"Over")))
=Original Spreadsheet!A29	=SUM(Lighting)	480	=B10-C10	=IF(D10>0,"Under",IF(D10=0,"Equal",IF(D10<0,"Over")))
=Original Spreadsheet!A33	=SUM(Sinks)	885	=B11-C11	=IF(D11>0,"Under",IF(D11=0,"Equal",IF(D11<0,"Over")))
=Original Spreadsheet!A36	=SUM(Toilets)	988	=B12-C12	=IF(D12>0,"Under",IF(D12=0,"Equal",IF(D12<0,"Over")))
=Original Spreadsheet!A39	=SUM(Walls)	1100	=B13-C13	=IF(D13>0,"Under",IF(D13=0,"Equal",IF(D13<0,"Over")))
Difference	=SUM(B5:B14)	=SUM(C5:C14)	=SUM(D5:D14)	
Incidentals			=Bathroom Remodel Summary!C16	
Job Loss/Job Covered			=D16+D15	
Highest Difference of Loss			=MIN(D5:D13)	

JOB 2: Formatted

Handy Handyman		
Remodeling Invoice		
Items	Material Cost	Labor Cost
Bath/Shower		
Tile	400.00	700.00
Grouting	50.00	100.00
Greenboard	100.00	150.00
Shower Pan	500.00	750.00
Floor Leveling	80.00	120.00
Bench Materials	350.00	500.00
Plumbing	200.00	600.00
Tub Removal	-	70.00
Cabinets		
Refinish and Paint	75.00	250.00
Countertops		
Granite	350.00	100.00
Faucets		
Showers	280.00	30.00
Sinks	1,200.00	320.00
Flooring		
Tile	380.00	425.00
Grout	114.00	220.00
Lighting		
Above Sinks	160.00	100.00
Ceiling Lights	80.00	180.00
Sinks		
Undermount	280.00	600.00
Toilets		
Installation	375.00	600.00
Walls		
Paint	175.00	760.00
Subtotal	5,149.00	6,575.00

Handy Handyman		
Remodeling Invoice		
Items	Total Cost	
Bath/Shower	4,670.00	
Cabinets	325.00	
Countertops	450.00	
Faucets	1,830.00	
Flooring	1,139.00	
Lighting	520.00	
Sinks	880.00	
Toilets	975.00	
Walls	935.00	
Labor and Materials		11,724.00
Incidentals	10%	1,172.40
Job Total		12,896.40
Down Payment	20%	2,579.28
Total Due		10,317.12

Handy Handyman				
Remodeling Invoice				
Items	Total Cost	Actual Cost	Difference	Over/Under
Bath/Shower	4,670.00	4,585.00	85.00	Under
Cabinets	325.00	630.00	(305.00)	Over
Countertops	450.00	450.00	-	Equal
Faucets	1,830.00	1,500.00	330.00	Under
Flooring	1,139.00	1,380.00	(241.00)	Over
Lighting	520.00	480.00	40.00	Under
Sinks	880.00	885.00	(5.00)	Over
Toilets	975.00	988.00	(13.00)	Over
Walls	935.00	1,100.00	(165.00)	Over
Difference	11,724.00	11,998.00	(274.00)	
Incidentals			1,172.40	
Job Loss/Job Covered			898.40	
Highest Difference of Loss			(305.00)	

JOB 3: Chart

